Campus Innovation Fund DAO Report

1. Introduction and Project Objective

For this project, we chose to develop a proof-of-concept decentralized autonomous organization (DAO) in the form of a Campus Innovation Fund DAO. The objective of this system is to create a structured process through which individuals can propose original ideas and potentially receive funding for them. More specifically, the goal of the DAO is to identify and support the strongest community ideas from a large pool of new project proposals.

We chose this project because funding allocation is a meaningful challenge in communities such as universities, student clubs, and research groups, where many promising ideas compete for limited resources. In many traditional funding systems, project selection can appear arbitrary or opaque. Decision-makers may not have access to the full set of proposals, individuals without formal authority may have limited influence over outcomes, and final decisions may depend too heavily on subjective judgment. In addition, the process itself is often slow and difficult to monitor.

At the beginning of this project, our aim was to build a DAO that could improve how this process is conducted. We wanted proposal submission, project evaluation, and funding decisions to follow a clear on-chain workflow rather than relying entirely on a small central group.

The solution we developed is a hybrid-governance funding DAO. In this system, members can submit project proposals, proposals can then be formally sponsored and voted on through a DAO governance process, and approved projects receive funding through milestone-based release rather than a single lump-sum payment. The design also includes a reputation mechanism so that governance influence depends not only on initial authority, but also on participation and project delivery outcomes over time.

2. High-Level System Overview

At a high level, the main participants in the DAO are members, proposers, voters, and the treasury. Members are registered participants in the system. A proposer is a member who submits a project idea for funding. Voters are the members who take part in governance decisions. The treasury is the part of the DAO that holds and releases funds for approved projects. In addition to these visible participants, the system also includes a formal governance contract that records proposals and voting outcomes, and a reputation mechanism that tracks member participation and project outcomes over time.

The DAO is organized into several layers. The governance layer manages formal proposal approval and voting. The token and reputation layer determines how voting power is formed by combining authority and earned reputation. The workflow layer manages project submission, project approval, milestone claims, and project status. The treasury layer controls the actual release of funds and applies treasury rules. Finally, the frontend layer provides a dashboard through which users can view proposals, projects, treasury status, and governance activity.

3. Example Workflow of the DAO

To elucidate the operational mechanics and governance protocols of the proposed Decentralized Autonomous Organization (DAO), this section presents a theoretical case study involving three network participants, denoted as Members A, B, and C. In this baseline scenario, all actors commence with a reputation score of zero. However, initial governance authority is asymmetric: only Member C possesses the requisite threshold of CIF governance tokens to formally instantiate an on-chain proposal. The workflow proceeds through four distinct operational phases:

Phase 1: Ideation and Structural Sponsorship

The process initiates when Member A submits a funding request for a campus sustainability initiative. The proposal outlines a total capital requirement of 0.2 WETH, structured to be disbursed across two equal milestones of 0.1 WETH. At this juncture, the submission exists strictly within the preliminary workflow layer as an "idea." Because Member A lacks the requisite CIF token threshold, they are structurally restricted from independently advancing the project to the formal Governor voting stage. Consequently, Member C must act as a formal sponsor. By utilizing their CIF authority to create the governance proposal, Member C bridges the gap between preliminary ideation and formal on-chain sponsorship.

Phase 2: Governance Voting and Ratification

Following formal sponsorship, the proposal enters the active voting epoch. In this instantiation, Members A and C cast affirmative votes, while Member B votes in the negative. Because the combined voting weight of A and C satisfies the protocol's threshold for approval, the proposal is ratified. Upon the conclusion of the voting period and the execution of necessary smart contract steps, the DAO's treasury formally earmarks the 0.2 WETH allocation.

Phase 3: Milestone-Based Treasury Disbursement

Crucially, project approval does not trigger an immediate, lump-sum capital release. Instead, the protocol mandates a milestone-contingent disbursement model to mitigate execution risk. The workflow proceeds as follows:

Milestone 1: Upon completing the initial phase of the initiative, Member A submits a verifiable claim. The DAO conducts a secondary governance review of this milestone. Upon approval, the treasury executes the first 0.1 WETH transfer.

Milestone 2: This procedural loop is replicated for the final deliverables. Member A submits a second claim, which is similarly reviewed and approved, prompting the release of the remaining 0.1 WETH. Following this transfer, the protocol registers the project as fully completed.

Phase 4: Dynamic Reputation Accrual

Concurrently, the system recalculates participant reputation to reflect historical engagement. While governance influence in the first round was exclusively predicated on static CIF token authority, subsequent epochs are influenced by behavioral metrics. Members who actively participated in the voting consensus receive reputation rewards. Furthermore, as the architect of a successfully executed project, Member A accrues significant performance-based reputation. Ultimately, this mechanism ensures that future governance power is dynamically weighted by a combination of baseline token authority and demonstrable, meritocratic contributions to the ecosystem.

4. Detailed Architecture and Mechanism Design

4.1 Governance Layer

The governance layer is responsible for formal proposal approval. It is built around an OpenZeppelin Governor framework and a timelock-controlled execution process. In this project, it provides the formal mechanism through which the DAO approves actions. A key design feature is the separation between idea submission and formal proposal sponsorship. A member may submit a project idea even if that member does not independently satisfy the threshold. A project only enters the Governor voting process once it is sponsored by a participant with sufficient governance authority.

The governance process follows a staged lifecycle. A formal proposal is created, the proposal enters a voting period, votes are collected, and if the proposal satisfies the passing conditions it is queued and executed through the timelock mechanism. This creates a distinction between a project existing in the workflow layer and a project being formally approved by governance. The same pattern is used both for initial project approval and for later milestone release decisions.

4.2 Token and Reputation Layer

The token and reputation layer determines how governance influence is formed. This layer combines a governance token with a non-transferable reputation system.

The governance token is the Campus Innovation Fund Token (CIF), which serves as the formal authority component of the DAO. CIF is implemented as an ERC20Votes token, so it supports both token balances and voting snapshots. In governance, token holding alone is not sufficient; voting power must also be delegated. As a result, CIF functions as the authority-based portion of voting power and also determines whether a participant can satisfy the proposal threshold required for formal sponsorship. From the implementation, CIF has the following core specifications:

- **Name:** Campus Innovation Fund Token
- **Symbol:** CIF
- **Decimals:** 18
- **Maximum Supply:** 1,000,000 CIF

The token contract includes a capped minting model, so new tokens cannot be minted beyond the maximum total supply. The current setup for the CIF is arbitrary at initiation. This is purposefully decided as different institutions adapting this DAO has distinct authority hierarchies. Some institutions can distribute CIF according to their positions by a higher authority. Others can trade authority through real-life contracts or trade-offs. In some cases, there may even be markets that operate based on CIF.

Alongside CIF, the DAO includes a separate ReputationRegistry. Reputation is not implemented as a tradeable token. Instead, it is stored as a non-transferable governance score associated with each registered member. This design allows the system to distinguish between assigned authority and earned participation. Reputation changes over time based on member behavior inside the DAO, including voting participation, milestone completion, and project success or failure. The final governance weight is produced by a HybridVotesAdapter, which combines the token and reputation components. The hybrid voting rule is defined as

$$V_i = V_i^{token} + V_i^{rep}$$

The token component is weighted at 60%:

$$V_i^{token} = 0.6 \times T_i$$

The reputation component is weighted at 40% and scaled relative to the active reputation of the DAO:

$$V_i^{rep} = 0.4 \times \left(\frac{R_i}{R_{total}} \right) \times S$$

where R_i is the reputation of member i, R_total is the total active reputation across all members, and S is the base voting supply used to keep the reputation component on the same scale as the token component.

This hybrid structure reflects the basic governance philosophy of the project: influence should depend partly on formal authority and partly on earned participation. In the example workflow, all members begin with zero reputation, so the first project approval is driven entirely by CIF authority. However, future governance becomes increasingly shaped by member behavior inside the DAO overtime.

4.3 Workflow and Funding Layer

The workflow layer is implemented primarily through FundingRegistry.SOL, which manages project submission, project approval state, milestone claims, and project status. This layer represents the operational logic of the DAO. At the proposal stage, a member submits a project with structured data including a title, a metadata reference, a recipient address, a requested funding amount, and a list of milestones. Each milestone has both a description and a funding amount. One important system rule is that the total requested funding must match the sum of milestone allocations exactly: **F = Σ f_m, for m = 1 to M**, where F is the total requested funding, M is the number of milestones, and f_m is the amount assigned to milestone m. This rule ensures that the full project budget is explicitly decomposed into measurable stages.

The workflow layer also stores proposals and project status transitions. A proposal moves through states such as submitted, voting, approved, or rejected. Once formally approved through governance, the proposal becomes an active project. The project then moves through milestone states as evidence is submitted and reviewed. A milestone cannot be released out of order; only the next unreleased milestone may be claimed.

This creates a clear sequential workflow and prevents arbitrary treasury withdrawals before earlier commitments have been completed. In operational terms, the workflow layer answers the following questions: Has a project been submitted? Has it been linked to a formal governance proposal? Has it been approved? Which milestone is currently claimable? Has a milestone claim been submitted? Has the project been completed?

4.4 Treasury Layer

The treasury layer is implemented through InnovationTreasury, which controls the release of project funding and applies financial rules to treasury management. The project budget is not released as a lump sum. Instead, the treasury recognizes an approved project budget and then releases WETH incrementally as milestones are approved. This is one of the key control mechanisms in the system. It means that governance approval gives a project access to a funding pathway, but not immediate access to the entire budget. The treasury also applies risk constraints. Two important rules are a maximum single-grant limit and a minimum liquid reserve requirement. The maximum single-grant rule prevents any one project from consuming too large a fraction of total treasury resources. In simplified form, the constraint can be written as $\mathbf{F \leq \alpha \times A}$. Where F is the approved project budget, A is the total managed treasury assets, and α is the maximum single-grant ratio. The reserve rule ensures that the treasury must retain a minimum liquid balance even when idle assets are supplied into external protocols. In simplified form: $\mathbf{L \geq \beta \times A}$, where L is liquid treasury balance and β is the reserve ratio. These rules make the treasury more than a passive wallet. It is an enforcement layer that ensures project funding decisions remain consistent with treasury safety constraints.

4.5 Oracle and DeFi Integration Layer

The DAO also includes an external integration layer. This consists of a TreasuryOracle, which reads price data from Chainlink, and an AaveWethAdapter, which allows idle treasury funds to be supplied into Aave. The oracle layer gives the system a way to estimate treasury value using current ETH/USD data. This allows the DAO to calculate net asset value and display treasury status in a more informative way than raw token balances alone.

The Aave integration serves a separate purpose. Not all treasury assets need to remain idle while projects are waiting for milestone release. The adapter allows part of the liquid WETH balance to be deposited into Aave and later withdrawn when funds are needed for approved project payouts. This introduces a simple treasury management feature into the DAO and connects governance decisions with broader DeFi infrastructure. In practice, this means the treasury layer manages both project disbursement and idle asset efficiency, while the oracle layer supports treasury visibility and valuation.

4.6 Reputation Update Logic

In the implemented prototype, reputation is updated through fixed event-based changes rather than through a nonlinear formula. Reputation evolves according to

$$R_i' = \min(1000, \max(0, R_i + \Delta))$$

where R_i is the current reputation of member i, Δ is the event-based change, and the score is bounded between 0 and 1000. The implemented events include rewards for governance participation and successful project delivery, as well as penalties for unsuccessful milestone claims.This means that reputation is endogenous to the DAO: members gain influence by participating in governance and by contributing to projects that are carried out successfully.

<u>The exact update logic is as follows</u>:

1. Voting on a project proposal

If a member participates in voting on a project proposal, then:

$$\boldsymbol{\Delta = +2}$$

2. Voting on a milestone-related proposal

If a member participates in voting on a milestone claim proposal, then:

$$\boldsymbol{\Delta = +2}$$

3. Milestone successfully released

If the proposer's milestone claim is approved and the milestone is released, then the proposer receives:

$$\boldsymbol{\Delta = +4}$$

4. Full project completion

If the final milestone is successfully completed and the project is marked complete, then the proposer receives an additional:

$$\boldsymbol{\Delta = +10}$$

5. Milestone rejection

If a milestone claim is rejected, then the proposer receives:

$$\boldsymbol{\Delta = -8}$$

Initial reputation

When a member is registered, the system assigns an initial reputation value. In the prototype deployment, the standard initialization used for members is:

$$\boldsymbol{R_i^{init} = 100}$$

4.7 Frontend Layer

The frontend layer provides the user-facing interface to the DAO. It does not replace the underlying contract logic, but it makes the system observable and usable. Through the dashboard, users can view proposal status, active projects, treasury balances, milestone progress, governance activity, and member information. The frontend is organized into dedicated views for proposals, projects, treasury state, and evidence outputs. In practical terms, it allows users to submit proposals, inspect governance status, monitor project progression, and verify treasury actions. This is especially important in a DAO context because the system is not only about executing rules, but also about making those rules visible and understandable to participants.

4.8 Integrated System Logic

Taken together, these layers form a sequential but interconnected system. The governance layer provides formal approval. The token and reputation layer determines influence. The workflow layer tracks the life cycle of projects and milestones. The treasury layer controls financial execution. The oracle and DeFi layer extends treasury visibility and idle fund management. The frontend layer exposes the state of the system to users.

5. Innovative Aspects and Design Extensions

The originality of this project comes from combining several governance ideas into one funding DAO rather than relying on a basic token-voting model.

5.1 Two-Stage Governance Pipeline

A key design choice is the separation between idea submission and formal proposal sponsorship. Members can introduce ideas into the system without immediately pushing them into full governance. A proposal only enters formal voting once it is sponsored by someone with sufficient authority. This creates a first level of screening and helps prevent the DAO from being overwhelmed by low-quality proposals. At the same time, it keeps the system open to broader participation at the idea stage.

5.2 Hybrid Voting: Authority and Reputation

A second important feature is the use of hybrid voting power. Instead of relying only on token ownership or only on reputation, the DAO combines both. CIF represents the authority side of governance, while reputation reflects participation and project performance. This design addresses a common weakness in standard token-based DAOs. If governance depends only on a token, voting power may reflect initial allocation more than actual contribution. At the same time, a system based only on reputation can be difficult to bootstrap and harder to integrate with standard governance frameworks. By combining the two, the project creates a compromise between formal authority and earned credibility. This allows the DAO to begin with a workable governance structure while still letting member behavior matter over time.

5.3 Reputation Logic

The reputation mechanism is designed to make governance somewhat self-reinforcing. In the current implementation, members build reputation by participating in votes and successfully delivering projects. This means that influence within the DAO is shaped not just by initial standing, but also by ongoing contributions.

In principle, reputation could also be adjusted relative to the total reputation in the system. As the DAO expands, high-reputation members would still carry more weight, but the scaling factor would increase slowly even as total participation rises. This helps prevent reputation from inflating too quickly, while still encouraging members to bring more participants into the system and support overall growth. We were not able to implement this nonlinear scaling in the final version. During development, we ran into technical challenges and ultimately fell back on a simpler approach using a bounded additive update rule.

5.4 Milestone-Based Accountability

A fourth innovative feature is milestone-based funding accountability. Approved projects do not receive their full budget immediately. Instead, funding is released step by step as milestones are claimed and approved. This makes the DAO more than a one-time voting system. Governance continues after approval, and treasury disbursement remains tied to project progress. In practice, this creates a stronger link between project delivery and funding release.

6. Additional Technical Features

In addition to the core governance workflow, the project includes several technical features that extend the DAO beyond a basic voting contract. These features include external data integration, DeFi treasury management, a navigable frontend interface, gas measurement and optimization, and a set of security and validation safeguards. Together, they make the system more complete as an applied smart-contract project.

6.1 Oracle Integration

The treasury system incorporates external market data through a Chainlink oracle. This oracle is used to read ETH/USD price information and convert treasury holdings into a more interpretable net asset value measure. As a result, the DAO is able to track treasury value not only in raw WETH terms, but also in USD terms. This feature is meaningful for two reasons. First, it connects the DAO to an external data source rather than keeping it entirely self-contained. Second, it improves treasury visibility by allowing project funding decisions and treasury state to be viewed in a broader financial context. The frontend and evidence outputs reflect this valuation logic by displaying treasury net asset value and the final treasury state after the governance scenario is executed.

6.2 DeFi Treasury Management Through Aave

The project also extends treasury functionality by integrating with Aave. Instead of treating the treasury as a passive wallet that only holds funds for future payouts, the DAO can place idle WETH into Aave and later withdraw it when milestone funding is needed. This introduces a basic but realistic treasury management feature into the system. In the demonstration scenario, one of the governance proposals is specifically used to deposit idle treasury funds into Aave, and a later proposal withdraws part of those funds before releasing a project milestone. This shows that the treasury is not only able to approve and release grants, but also able to manage unused capital more actively between funding events.

6.3 Frontend Interface and Navigation

The project includes a frontend interface with multiple views rather than a single static page. The frontend allows users to move between proposal pages, project-detail pages, treasury views, milestone-claim forms, and evidence outputs. This makes the DAO easier to inspect and interact with from a user perspective. The interface is important because the system is not only about contract execution, but also about making governance state understandable to participants. A user can trace proposal status, project progress, treasury balances, and milestone activity without needing to inspect raw contract state directly. This makes the project more practical as a governance application rather than only a contract demonstration.

6.4 Governance Token Meaning and Value

Another important feature of the project is the way the governance token is tied to authority. The CIF token is not treated as a token with arbitrary symbolic value. Instead, it is directly linked to governance power inside the DAO. It affects formal sponsorship, voting power, and the ability to influence project approval. This gives the token a defined role within the system.

Because authority inside the DAO is attached to CIF, the token acquires institutional meaning rather than existing only as a speculative asset. In some contexts, this can also support market value, since the token represents access to governance influence within a functioning project-funding system. In that sense, the token's value is connected not only to ownership, but also to the practical role it plays in decision-making.

6.5 Gas Measurement and Optimization

The project also looked closely at gas usage and how to optimize it. We tracked gas costs across key governance and treasury actions, such as creating proposals, voting, queueing, execution, treasury approvals, milestone releases, and deposits and withdrawals. This helped us clearly see which parts of contract execution were more expensive and where we could improve efficiency. As part of this analysis, we implemented a specific optimization in the treasury logic. During development, we updated part of the treasury flow so that known balance changes could be reused, instead of triggering an extra adapter read each time. This change directly reduced the gas cost of deposit transactions. Overall, this shows that gas efficiency was not treated as an afterthought. It was considered throughout the development process, with practical adjustments made along the way to improve performance and reduce costs.

6.6 Security and Validation Guardrails

Beyond the visible governance logic, the project includes several safeguards designed to improve contract safety and package integrity. The treasury and Aave adapter are protected against reentrancy during sensitive actions such as milestone payout, deposit, and withdrawal. The oracle layer also checks for stale or invalid price data before valuation is used. These controls reduce the risk of unsafe treasury execution or silent dependence on unreliable external data.

The project also includes validation work outside the contracts themselves. Static-analysis outputs, submission-package checks, repository scans, and deliverable validation scripts were all used during development. These measures do not change the DAO's governance logic directly, but they do improve the quality of the final system by making the codebase more auditable and the submission package more consistent.

7. Testing, Validations, and Results

The project includes one full recorded live scenario on Sepolia using deployed contract addresses. Rather than only showing isolated contract execution, this scenario demonstrates the DAO operating as a complete system. The recorded sequence contains three formal governance proposal events: project approval, treasury capital management through Aave, and milestone-based payout release.

7.1 Project Approval Event

The first proposal is the project-approval event. In this step, the funding proposal is formally approved through governance and the project becomes active in the system. This confirms that the submission and sponsorship logic can successfully lead to a valid governance outcome and an approved project state. The recorded lifecycle view shows that the proposal progressed through the expected governance stages, and the resulting project-state view shows that the project budget was recognized after approval.

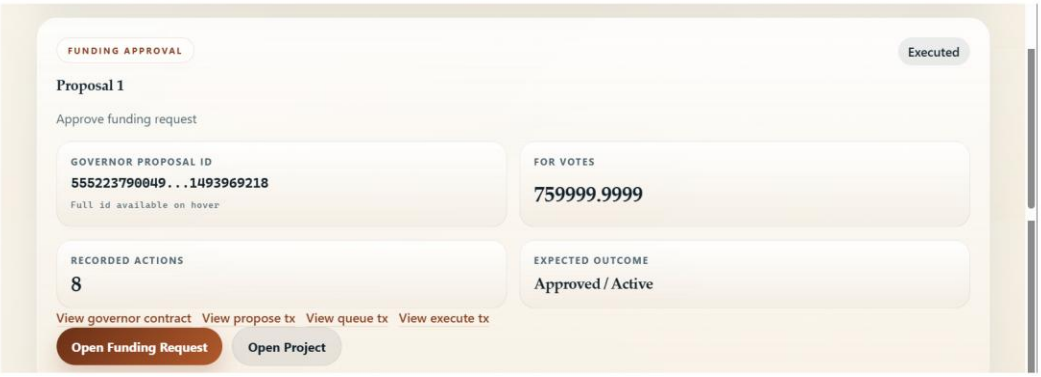

Figure 1. Proposal 1 lifecycle after formal approval.

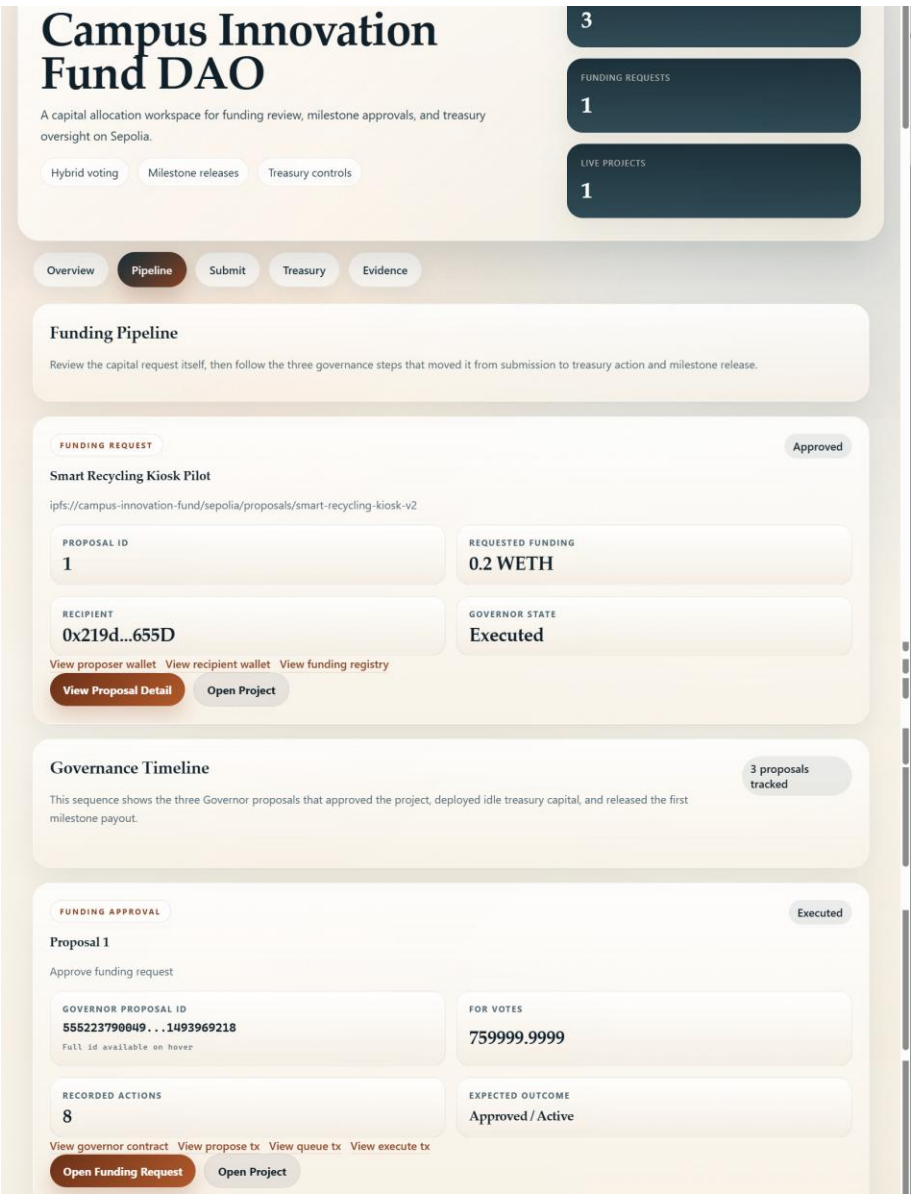

Figure 2. Project state after Proposal 1 is executed.

7.2 Treasury Management Event

The second proposal is the treasury-management event. In this step, idle WETH held by the treasury is deposited into Aave. This shows that the DAO is capable of governing not only project selection, but also treasury management decisions involving an external DeFi protocol. The recorded state after this proposal shows that treasury funds were moved into Aave as intended, which demonstrates that the treasury layer can respond to governance decisions beyond grant approval alone.

Figure 3. Treasury state after Proposal 2 deposits idle funds into Aave.

7.3 Milestone Payout Event

The third proposal is the milestone-payout event. In this step, the treasury withdraws the required amount of funds and releases the first milestone payment to the approved project. This demonstrates that project approval does not lead to an immediate full payout. Instead, funding is released in stages after milestone-related governance actions are completed. The recorded results show both the payout event itself and the change in the recipient's balance after the milestone release.

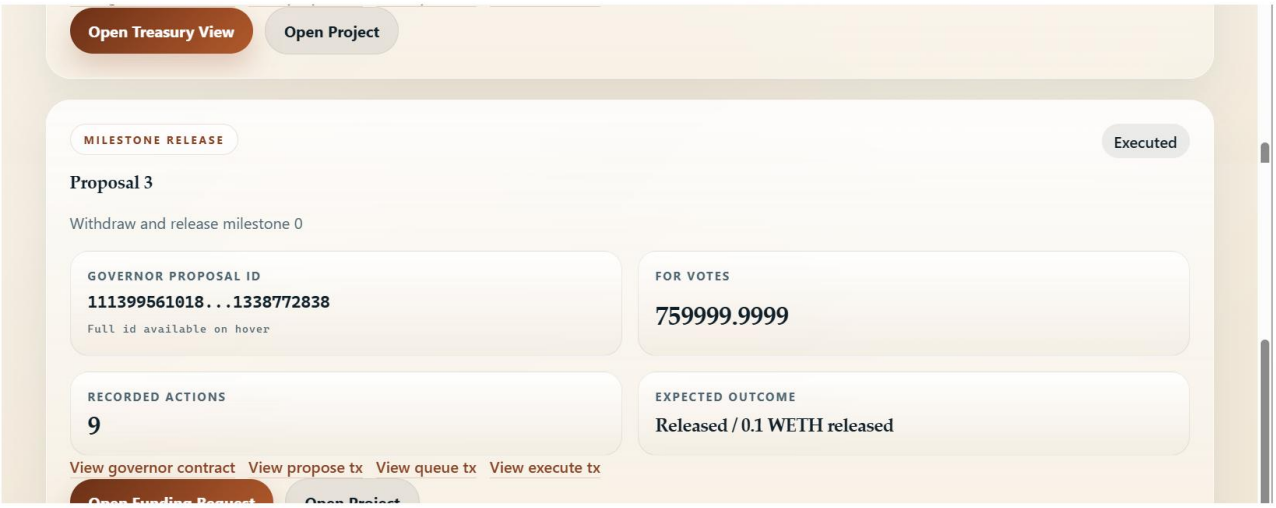

Figure 4. Proposal 3 payout and milestone-release state.

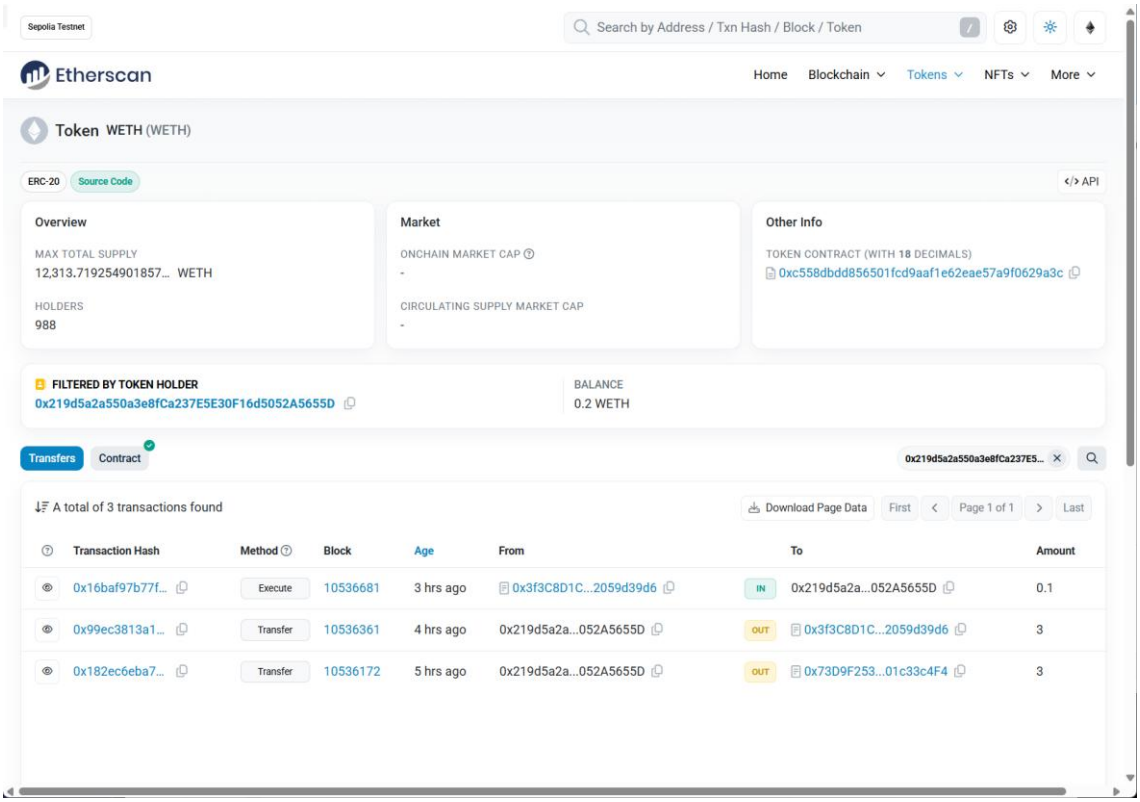

Figure 5. Recipient balance before and after the milestone payout.

7.4 Final Recorded State

Taken together, the three proposal events form one complete live governance scenario: first, a project is approved; second, treasury capital is managed through Aave; third, milestone-based funding is released. The final treasury view reflects the resulting net asset value and asset position after these actions.

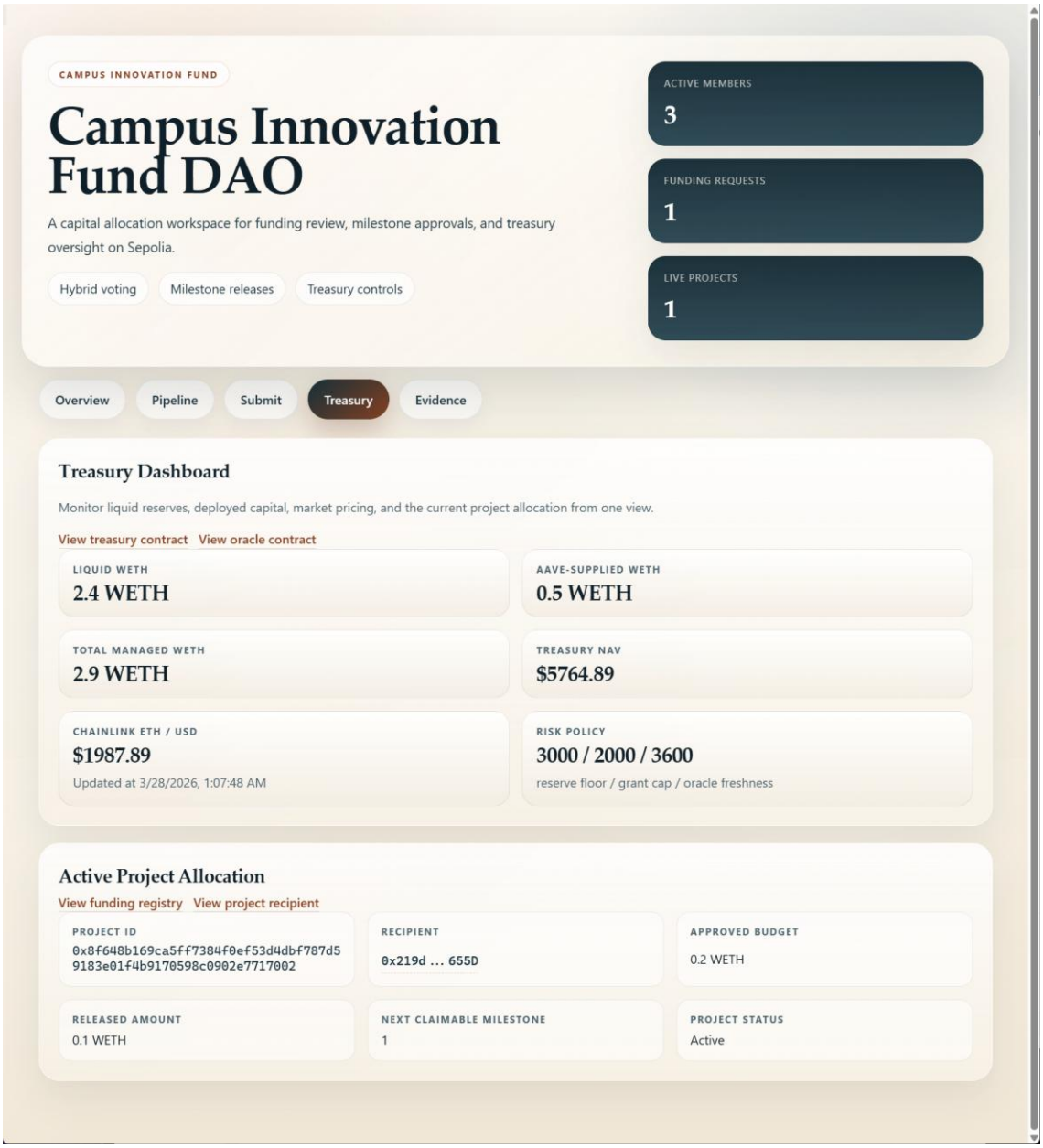

Figure 6. Final treasury state after the recorded Sepolia scenario.

7.5 What the Results Show

The results we recorded match what the system was designed to do. The project moved smoothly from submission to approval, idle treasury funds were deposited into Aave, and the first milestone was funded through a governance-approved payout. In other words, the DAO worked as expected in a real, end-to-end scenario. Governance decisions triggered the right workflow steps, treasury actions happened in the correct sequence, and milestone-based payments were executed as designed.

8. Business Use Case and Practical Relevance

8.1 Innovation Funding

A clear use case for this DAO is funding innovation in environments where many early-stage ideas compete for limited resources. This includes universities, student innovation programs, research groups, and nonprofit grant initiatives. In these settings, there are often more strong ideas than available funding, and decisions may rely on small committees, informal networks, or limited review capacity. This DAO framework introduces a more structured and transparent process. Proposals are submitted through a formal pipeline, supported by sponsors, and evaluated through decentralized voting. At the same time, it addresses the execution risk that comes with early-stage projects by linking funding to milestone completion. Instead of releasing all funds upfront, the treasury distributes capital in phases, which helps reduce the risk of misallocation while still supporting high-potential but uncertain ideas.

If desired, the reputation component in this type of system could also be represented through a soulbound-style non-transferable token rather than only a registry-based score, giving reputation a more explicit on-chain credential form while keeping it non-transferable.

8.2 Decentralizing Authority in Hierarchical Membership Organizations

Traditional membership organizations, such as student associations or professional clubs, often suffer from rigid authority structures where decision-making power is strictly tethered to formal officer roles or legacy status. This positional concentration of authority impedes organizational agility and frequently disenfranchises highly engaged, newer participants. The DAO architecture resolves this structural rigidity by deploying the CIF governance token to represent formal voting rights, intricately balanced by a dynamic reputation system that tracks actual project delivery and active participation. Consequently, the model fosters a meritocratic governance environment where organizational influence shifts fluidly based on demonstrable value creation, rather than remaining locked in static hierarchical positions.

8.3 Participatory Budgeting and First-Stage Screening

A persistent challenge in decentralized governance and municipal participatory budgeting is mitigating voter fatigue and maintaining agenda efficiency when faced with a high volume of raw submissions. The DAO addresses this via a bifurcated, two-stage mechanism design that explicitly separates initial idea submission from formal proposal sponsorship. This structural delineation mirrors best practices in public resource allocation: broad, frictionless participation is encouraged during the ideation phase, but a rigorous intermediate screening protocol is required before advancing to a final vote. By mandating formal sponsorship as a gating mechanism, the system effectively shields the governance layer from informational overload and low-quality proposals, optimizing voter attention and elevating the overall caliber of organizational decision-making.

8.4 Why This Governance Structure Is Useful

Across these three settings, the DAO is useful because it combines several functions that are often separated in practice. It opens proposal access more broadly, preserves structure in agenda-setting, combines formal authority with earned participation, and ties funding release to performance over time.

For organizations that want a decision system that is more participatory than a small committee but more disciplined than unrestricted open voting, this DAO provides a practical model. Its structure is flexible enough to apply to innovation grants, membership-based organizations, and public or community budgeting, while still keeping governance, funding, and accountability linked in one system.

9. Conclusion, Implementation Summary, and Lessons Learned

This project developed a proof-of-concept Campus Innovation Fund DAO designed to support project funding through a structured governance process. The final system combines proposal submission, formal sponsorship, hybrid voting, milestone-based treasury release, and reputation updates within one DAO framework. Rather than treating governance as a single approval event, the project models funding as a staged process in which projects are proposed, reviewed, approved, monitored, and funded over time.

Major Steps

- **Step 1: Designing the DAO architecture.** The first and most important step was building the conceptual structure of the DAO. From OpenZeppelin, we adopted the idea of a formal governance layer. From Gitcoin-related governance models, we drew the distinction between authority and reputation as separate sources of influence. From the broader history of Bitcoin and decentralized systems, we developed the idea that the reputation component of governance can be self-reinforcing, so that its value increases as the DAO itself grows. Finally, drawing on our financial background, we added a treasury-management component in which unused funds could be placed into Aave rather than remaining idle.
- **Step 2: Coding and implementation.** The second major step was implementation. This involved converting the conceptual design into a working smart-contract system, along with the related frontend and treasury features. Some of this coding work was assisted by AI-based coding tools and was primarily handled by my teammate.
- **Step 3: Testing and writing the report.** The final major step was validating the project through testing and presenting it clearly in written form. We designed a live scenario and a sequence of governance proposals that would exercise the major parts of the DAO, including project approval, treasury capital management, and milestone-based funding release. At the same time, we organized the project into a report that explains the logic, structure, and practical relevance of the system in a clear way.

Main Lessons Learned

- **Lesson 1: The importance of efficiency and transparency in DAO design.** Through this project, we developed a stronger understanding of why DAOs can be useful for real-world organizational problems. A well-designed DAO can make proposal handling, voting, and fund release more structured and more visible than a conventional process, especially in settings where many ideas compete for limited funding.
- **Lesson 2: The importance of precise prompting and system definition when using AI for coding.** We learned that AI tools are most useful when the design is already clearly specified. In practice, this meant defining inputs, outputs, variables, and contract responsibilities carefully before asking for implementation help. More detailed prompts led to better and more reliable results.
- **Lesson 3: The importance of dividing teamwork by function rather than by disconnected code fragments.** At first, we considered dividing the assignment by coding tasks. However, we found that synchronizing different parts of the system was too difficult when inputs, outputs, and shared state were tightly connected. As a result, we changed our approach and divided the work into information gathering and idea generation, actual implementation and testing, and report generation. This turned out to be a more effective way to manage a complex assignment.